

What is the concordance between preoperative molecular testing results (ThyroSeq vs Afirma), Bethesda score, and final histopathologic diagnosis in patients with 2-4 cm thyroid nodules undergoing thyroidectomy?

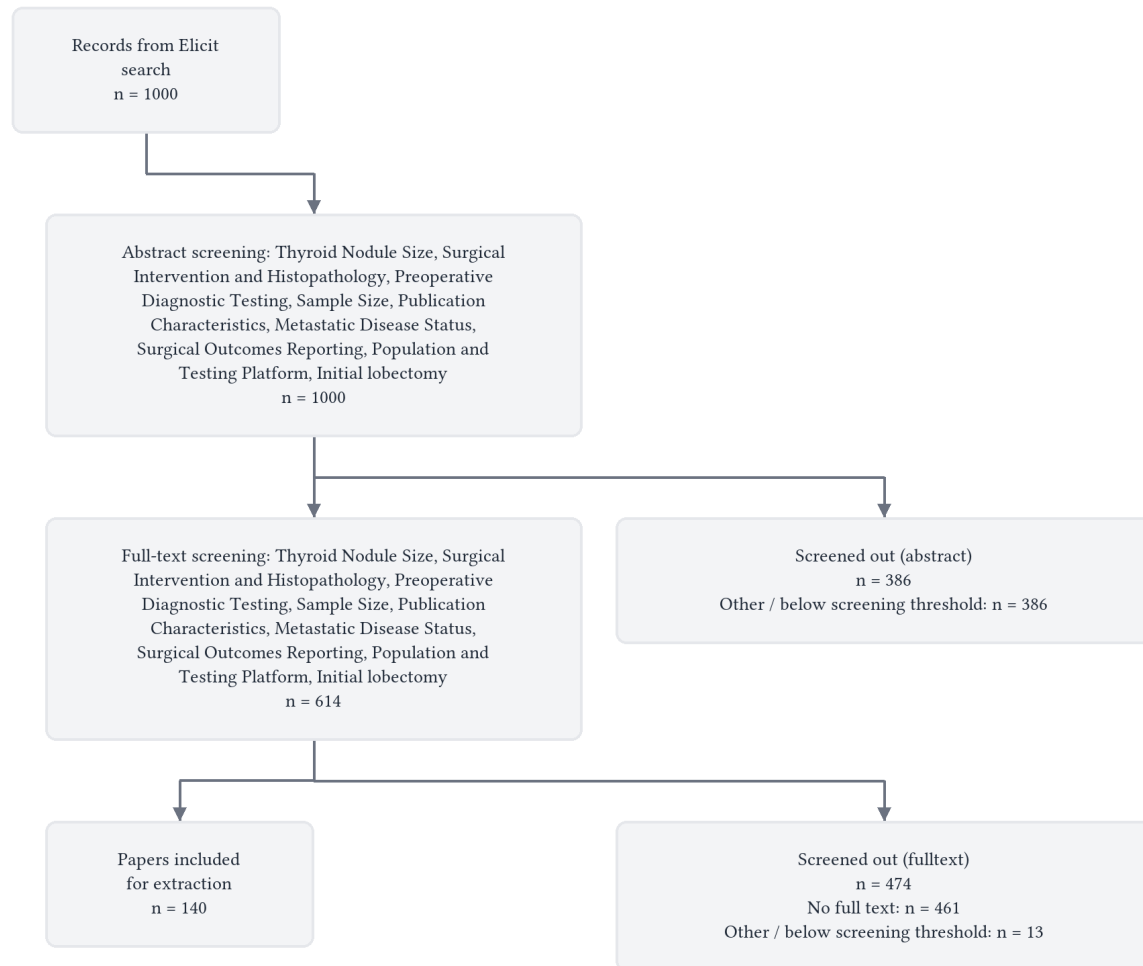
While no studies specifically examined the 2-4 cm nodule cohort, available evidence from studies including nodules in this size range demonstrates that both ThyroSeq and Afirma achieve >90% concordance with histopathology for ruling out malignancy in Bethesda III-IV nodules, though ThyroSeq shows higher specificity (82-91% vs. 66-88%) and both platforms have variable positive predictive values (35-77%) due to inclusion of NIFTP and low-risk cancers among molecularly-positive results.

Abstract

This systematic review of 80 studies examining molecular testing in thyroid nodules reveals limited data specifically addressing the 2-4 cm size range, though several studies included nodules within this range as part of broader cohorts with median sizes of 1.8-2.8 cm [1-4]. For indeterminate thyroid nodules (Bethesda III and IV), molecular testing significantly improved concordance with final histopathology compared to cytology alone, with optimal surgical management achieved in 67.5% of patients undergoing molecular testing versus 25.1% without testing ($p<0.001$) [5]. ThyroSeq v3 demonstrated sensitivity of 94% [3], specificity of 82% [3], and NPV of 97% [3], while Afirma GSC showed real-world sensitivity of 97% [6], specificity of 88% [6], and NPV of 99% [6]. In the only head-to-head randomized comparison, ThyroSeq v2 achieved higher specificity (91% vs. 66%) and PPV (57% vs. 39%) than Afirma GEC, enabling more patients to avoid surgery (62% vs. 39%) [4].

Bethesda III malignancy rates ranged from 9.2% [7] to 77.7% [8] and Bethesda IV from 22.8% [7] to 76.4% [8], with variation explained by surgical selection bias and institutional practice patterns [8, 9]. Both platforms demonstrated >90% NPV for ruling out malignancy [3, 6], though PPV remained more variable (35-77%) [4, 10] due to inclusion of NIFTP and low-risk cancers among molecularly-positive results. Platform performance differed by tumor subtype: BRAF V600E showed 100% concordance with malignancy [11] but only 40% sensitivity for follicular variant PTC [12], while both platforms showed reduced performance in oncocytic neoplasms (20% PPV) [13]. Critical evidence gaps include lack of size-specific analyses for 2-4 cm nodules, limited head-to-head platform comparisons, insufficient long-term surveillance data for test-negative results, and absent completion thyroidectomy rates stratified by molecular testing.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What is the concordance between preoperative molecular testing results (ThyroSeq vs Afirma), Bethesda score, and final histopathologic diagnosis in patients with 2-4 cm thyroid nodules undergoing thyroidectomy?"

The search returned 1,000 total results from Elicit.

We retrieved 1000 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Thyroid Nodule Size:** Does the study include patients with thyroid nodules measuring 2-4 cm in diameter?
- **Surgical Intervention and Histopathology:** Does the study include patients who underwent thyroidectomy (lobectomy or total thyroidectomy) with final histopathologic diagnosis reported?
- **Preoperative Diagnostic Testing:** Does the study report Bethesda cytology categories and/or preoperative molecular testing results using current platforms (ThyroSeq v3/v4, Afirma GSC, or equivalent)?
- **Sample Size:** Does the study include a cohort of 50 or more patients?
- **Publication Characteristics:** Is this an English-language study published between 2015 and March 2026?
- **Metastatic Disease Status:** Does the study exclude patients with preoperatively clinically evident lymph node metastases or distant metastases (or focus only on patients without evident metastases)?
- **Surgical Outcomes Reporting:** Does the study report surgical outcomes such as extent of initial surgery (lobectomy vs total thyroidectomy) and/or completion thyroidectomy rates?
- **Population and Testing Platform:** Does the study focus on adult populations (not purely pediatric <18 years) and use current molecular testing platforms (not outdated platforms like Afirma GEC or ThyroSeq v1/v2)?
- **Initial lobectomy**

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

614 papers passed abstract screening and moved to full-text screening.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 386

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Thyroid Nodule Size:** Does the study include patients with thyroid nodules measuring 2-4 cm in diameter?
- **Surgical Intervention and Histopathology:** Does the study include patients who underwent thyroidectomy (lobectomy or total thyroidectomy) with final histopathologic diagnosis reported?
- **Preoperative Diagnostic Testing:** Does the study report Bethesda cytology categories and/or preoperative molecular testing results using current platforms (ThyroSeq v3/v4, Afirma GSC, or equivalent)?
- **Sample Size:** Does the study include a cohort of 50 or more patients?
- **Publication Characteristics:** Is this an English-language study published between 2015 and March 2026?
- **Metastatic Disease Status:** Does the study exclude patients with preoperatively clinically evident lymph node metastases or distant metastases (or focus only on patients without evident metastases)?
- **Surgical Outcomes Reporting:** Does the study report surgical outcomes such as extent of initial surgery (lobectomy vs total thyroidectomy) and/or completion thyroidectomy rates?
- **Population and Testing Platform:** Does the study focus on adult populations (not purely pediatric <18 years) and use current molecular testing platforms (not outdated platforms like Afirma GEC or ThyroSeq v1/v2)?
- **Initial lobectomy**

We considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

140 papers passed full-text screening and moved to data extraction.

At full-text screening, the number of papers excluded for each primary reason was:

- **No full text:** n = 461
- **Other / below screening threshold:** n = 13

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Population:**

Extract study population details specifically for patients with 2-4 cm thyroid nodules undergoing thyroidectomy, including:

- Total sample size in the 2-4 cm cohort
- Mean/median nodule size and confirmation of 2-4 cm range
- Patient demographics (age, sex distribution)
- Bethesda category distribution in this size cohort
- Inclusion/exclusion criteria relevant to nodule size
- Any subgroup analysis by nodule size within the 2-4 cm range

- **Molecular Testing Details:**

Extract comprehensive details about preoperative molecular testing platforms used, including:

- Specific platform(s): ThyroSeq (version), Afirma GSC, or other
- Result classification system used (e.g., positive/negative, suspicious/benign, risk categories)
- Number of patients tested with each platform
- Technical success rate of molecular testing
- Any head-to-head comparisons between ThyroSeq and Afirma in the same patients
- Timing of molecular testing relative to surgery

- **Bethesda Categories:**

Extract Bethesda cytology scoring details for the 2-4 cm nodule cohort, including:

- Distribution across Bethesda categories (I-VI)
- Focus on indeterminate categories (III, IV, V) as primary interest
- Any reclassification or repeat FNA results
- Concordance between initial and repeat cytology when available
- Cytopathologist experience or inter-observer variability data

- **Final Histopathology:**

Extract final histopathologic diagnosis details for 2-4 cm nodules, including:

- Specific histologic diagnosis (benign, malignant subtypes)
- Cancer prevalence in the cohort

- Distribution of cancer types (papillary, follicular, medullary, etc.)
- NIFTP (noninvasive follicular thyroid neoplasm with papillary-like nuclear features) classification
- Aggressive features when present (invasion, high-grade, etc.)
- Any cases requiring central pathology review or consensus

- **Molecular-Histology Concordance:**

Extract concordance measures between preoperative molecular testing results and final histopathologic diagnosis, including:

- Sensitivity, specificity, PPV, NPV of molecular testing
- Overall accuracy percentage
- Kappa statistic or other agreement measures
- False positive and false negative rates
- Specific breakdown by molecular platform (ThyroSeq vs Afirma)
- Performance differences by Bethesda category
- Concordance for detecting aggressive vs indolent malignancies

- **Bethesda-Histology Concordance:**

Extract concordance measures between Bethesda cytology categories and final histopathologic diagnosis for 2-4 cm nodules, including:

- Malignancy rate by Bethesda category (especially III, IV, V)
- Sensitivity and specificity of cytology alone
- Agreement statistics between cytologic and histologic diagnosis
- Rate of upgrade/downgrade from cytology to final pathology
- Performance of cytology in detecting specific cancer subtypes

- **Three-Way Concordance Analysis:**

Extract any analysis of concordance between all three diagnostic modalities (molecular testing, Bethesda score, and histopathology) in 2-4 cm nodules, including:

- Cases with agreement across all three modalities
- Discordant cases and patterns of disagreement
- Added value of molecular testing beyond cytology alone
- Decision trees or algorithms combining molecular and cytologic results
- Clinical scenarios where molecular testing changed management based on cytology
- Subgroup analyses by specific molecular alterations or Bethesda categories

- **Platform Comparison:**

Extract direct comparisons between ThyroSeq and Afirma platforms when both are reported or compared, including:

- Head-to-head performance metrics in the same patient cohorts
- Differences in sensitivity, specificity, PPV, NPV between platforms
- Concordance between ThyroSeq and Afirma results when both tested
- Platform-specific performance by Bethesda category
- Cost-effectiveness or practical implementation differences mentioned
- Any preference or recommendation for one platform over another

- Platform-specific correlation with specific genetic mutations or cancer subtypes
- **Study Methodology:**
Extract methodological details that affect validity of concordance analyses, including:
 - Study design (prospective, retrospective, cohort, case-control)
 - Blinding of molecular testing, cytology, and histology interpretations
 - Central vs local pathology review processes
 - Statistical methods used for concordance analysis
 - Sample size calculations or power analysis for concordance studies
 - Handling of indeterminate or non-diagnostic molecular test results
 - Potential sources of bias acknowledged by authors

Report

Due to the limitations of the AI model, we are only able to process 80 sources while writing a report. This report was written using the 80 sources that had the highest screening scores out of the 140 sources that we screened in and extracted data from.

Results

Characteristics of Included Studies

The systematic review identified 80 studies examining concordance between preoperative molecular testing, Bethesda cytology scoring, and histopathologic diagnosis in thyroid nodules. The majority of studies were retrospective cohort designs [1, 2, 5], with fewer prospective studies [3, 14, 15]. Most studies did not explicitly report on the 2-4 cm nodule size range specifically, though several included nodules within this range as part of broader cohorts.

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
Alexandra E. Payne et al., 2025 [5]	Yes	Retrospective cohort [5]	410 patients	Not specified for 2-4 cm	Mean age 52.8 years, 81.2% female [5]	Bethesda III: 246, Bethesda IV: 164 [5]
Jessica Hier et al., 2021 [1]	Yes	Retrospective chart review [1]	122 patients	Control: 1.9 cm (1.5-2.2), Molecular: 1.8 cm (1.6-2.0) [1]	Control: 54.1 years, Molecular: 49.2 years [1]	Bethesda V and VI [1]
J. Lukáš et al., 2023 [14]	Yes	Prospective retrospective analysis [14]	287 patients, 52 indeterminate	Not specified	Not specified	Bethesda III: 51.9%, Bethesda IV: 48.0% [14]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
Mehmet Kostek et al., 2025 [2]	Yes	Retrospective [2]	177 patients	Median 2.8 cm (IQR: 1.95-4.0) [2]	Mean age 55.9 years, 79.7% female [2]	Bethesda III: 86.4%, Bethesda IV: 13.6% [2]
Allan Golding et al., 2025 [16]	Yes	Retrospective cohort [16]	Training: 697, Validation: 259	Not specified	Not specified	Training: 55% BIII/IV, 45% BV/VI [16]
Elizabeth J. de Koster et al., 2023 [17]	Yes	Prospective multicenter RCT [17]	132 patients, 115 with molecular testing	Not specified for 2-4 cm	Not specified	Bethesda III: 45%, Bethesda IV: 55% [17]
D. Steward et al., 2018 [3]	Yes	Prospective blinded cohort [3]	286 nodules, 257 informative	Median 2.4 cm [3]	Median age 53 years, 79% female [3]	Bethesda III: 172, Bethesda IV: 101, Bethesda V: 13 [3]
Alexandra E. Payne et al., 2024 [18]	Yes	Multicenter retrospective [18]	203 patients	Bethesda III: 2.17 cm, Bethesda IV: 2.27 cm [18]	Bethesda III: 54.4 years, Bethesda IV: 51.9 years [18]	Bethesda III: 117, Bethesda IV: 86 [18]
Roberto Olmos et al., 2023 [19]	Yes	Prospective noninterventional [19]	155 with molecular testing	Median 1.6 cm [19]	Not specified	Bethesda III: 56.1%, Bethesda IV: 43.9% [19]
J. Sistrunk et al., 2019 [20]	Yes	Retrospective observational [20]	337 patients	Median 2.1 cm (0.6-7.0) [20]	Median age 61 years, 69% female [20]	Bethesda III: 69%, Bethesda IV: 31% [20]
Max A. Schumm et al., 2021 [21]	Yes	Prospective longitudinal [21]	174 patients (252 eligible)	Median 2.0 cm [21]	Median age 56 years, 77% female [21]	Bethesda 3: 86.2%, Bethesda 4: 13.8% [21]
Grégoire B. Morand et al., 2025 [22]	Yes	Retrospective multicenter [22]	718 patients	Median 15 mm (IQR 10-24) [22]	Mean age 50.6 years, 77.4% female [22]	III: 20.3%, IV: 20.3%, V: 26.0%, VI: 33.3% [22]
D'Addino José Luis et al., 2025 [23]	Yes	Retrospective [23]	88 patients	Median 2 cm, mean 2.47 cm [23]	Mean age 58 years, 84.1% female [23]	I: 3.6%, II: 66.1%, III: 3.6%, IV: 12.5%, V: 5.4%, VI: 8.9% [23]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
Rachel C Jug et al., 2018 [24]	Yes	Retrospective [24]	Not specified for 2-4 cm	Not specified	Not specified	Majority indeterminate (III and IV) [24]
Alicia Belaiche et al., 2024 [25]	Yes	Retrospective chart review [25]	50 patients (27 FTC, 23 OTC)	FTC: 3.0 cm, OTC: 3.5 cm [25]	FTC: 51.1 years, OTC: 61.8 years [25]	FTC III: 29.6%, IV: 59.3%; OTC III: 34.8%, IV: 52.2% [25]
Brandon R. Rosvall et al., 2021 [15]	Yes	Prospective [15]	124 patients	Not specified; 76.6% between 1.0-3.9 cm [15]	86.3% female, 51.6% ≥55 years [15]	I: 22.6%, II: 41.1%, III: 7.3%, IV: 6.5%, V: 4%, VI: 18.5% [15]
Noha Elsherbini et al., 2022 [26]	Yes	Retrospective [26]	621 molecular profiling	Average size 2.7 cm (SD 1.3) [26]	Mean age 62 years, 81% female [26]	Bethesda III: 68%, Bethesda IV: 32% [26]
J. Pardal-Refoyo et al., 2025 [27]	Yes	Systematic review [27]	Meta-analysis of multiple studies	Not specified	Not specified	Focus on Bethesda III and IV [27]
Elaine Y F Huang et al., 2023 [28]	Yes	Retrospective cohort [28]	446 patients, 630 nodules	Not specified	Age 24-89 years, mean 60; 83% female [28]	Most Bethesda 2 (78.1%) [28]
E. Kandil et al., 2023 [29]	Yes	Retrospective [29]	408 patients total	Not specified	Not specified	Predominantly Bethesda III [29]
R. Xia et al., 2021 [30]	Yes	Retrospective cohort [30]	81 cases analyzed	Not specified	Not specified	Bethesda III: 79.0% [30]
S. Chumber et al., 2023 [31]	Yes	Prospective observational [31]	80 patients	Not specified	Not specified	II: 38.75%, III: 23.75%, IV: 18.70%, V: 1.25%, VI: 11.25% [31]
Raisa Chowdhury et al., 2025 [32]	Yes	Systematic review [32]	Meta-analysis	Not specified	Not specified	Bethesda III: 46-85%, IV: 15-42% [32]
G. Penna et al., 2021 [33]	Yes	Prospective cohort [33]	62 patients	Not specified	Mean age 45.27 years [33]	Bethesda III and IV focus [33]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
M. Capezzone et al., 2025 [34]	Yes	Prospective [34]	104 patients	Median 21 mm [34]	Median 59 years, 72 females, 32 males [34]	TIR3A: 69.5%, TIR3B: 30.5% [34]
M. Kaško et al., 2023 [11]	Yes	Cross-sectional [11]	170 patients	Not specified	Not specified	III: 98, IV: 35, V: 37 [11]
L. Hargitai et al., 2020 [35]	Yes	Retrospective cohort [35]	81 Bethesda IV/V	Not specified	Bethesda IV: 46 female, 14 male; V: 16 female, 5 male [35]	IV: 60, V: 21 [35]
M. Rossi et al., 2015 [36]	Yes	Prospective [36]	911 patients, 940 nodules	Not specified	Mean age 48 years, sex ratio 3.4:1 [36]	140 indeterminate lesions [36]
G. Lithwick-Yanai et al., 2016 [37]	Yes	Blinded multicenter retrospective [37]	201 validation samples	Not specified	Not specified	Bethesda III, IV, V focus [37]
C. Gomes-Lima et al., 2020 [38]	Yes	Retrospective cohort [38]	47 patients	Not specified	Median age 51 years [38]	Bethesda III and IV [38]
I. Lončar et al., 2023 [13]	Yes	Prospective multicenter [13]	68 nodules in 66 patients	Mean 30.3 mm [13]	Mean age 52.5 years, 71.2% female [13]	III: 39.7%, IV: 60.3% [13]
S. Thiagarajan et al., 2024 [8]	Yes	Retrospective cohort [8]	359 patients	Not specified	Majority female (81.8%) [8]	Bethesda 3 and 4 focus [8]
Julián Gómez Herrera et al., 2026 [39]	Yes	Retrospective observational [39]	164 patients	Not specified	Not specified	III: 49.4%, IV: 50.6% [39]
Q. Yu et al., 2026 [12]	Yes	Retrospective [12]	180 Bethesda III	Not specified	Not specified	Focus on Bethesda III [12]
G. Fu et al., 2024 [40]	Yes	Prospective diagnostic [40]	249 FNA biopsies	Not specified	Female: 75.2%, male: 24.8% [40]	ND: 30.5%, Indeterminate: 50.6%, Malignant: 13.7%, Benign: 5.2% [40]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
M. Molina-Vega et al., 2019 [41]	Yes	Retrospective multicenter [41]	2809 nodules	Mean 21.8 ± 12.3 mm; 40.1% in 20-40 mm range [41]	Mean age 51.9 ± 14.3 years, 82.7% female [41]	II: 81.2%, III: 6.1%, IV: 5.3%, V: 0.9%, VI: 2.2% [41]
Iyad Hassan et al., 2024 [42]	Yes	Retrospective cohort [42]	1038 patients	Not specified	Not specified	II: 26.9%, III: 25.4%, IV: 11.5%, V: 19.1%, VI: 12.1% [42]
L. Teodoriu et al., 2024 [43]	Yes	Prospective noninterventional [43]	44 patients (MIBI cohort)	Not specified	Median age 57 years [43]	I: 15, II: 27, III: 5, V: 2 [43]
Mehmet Üstün et al., 2023 [44]	Yes	Retrospective [44]	72 patients	Category 3: 23.03 mm, Category 4: 24.59 mm [44]	Not specified	Category 3: 40, Category 4: 32 [44]
Li-Cheng Tan et al., 2021 [45]	Yes	Retrospective [45]	73 patients	Not specified	Not specified	II: 14, III/IV: 27, V/VI: 32 [45]
Y. Nikiforov et al., 2015 [10]	Yes	Prospective [10]	465 AUS/FLUS nodules	Not specified	Not specified	AUS/FLUS (Bethesda III) [10]
Rupendra T. Shrestha et al., 2016 [46]	Yes	Retrospective [46]	261 operated nodules	Not specified	Mean age 48 years, sex ratio 3.4:1 [46]	AUS/FLUS: 28%, SFN: 11%, SUSP: 9% [46]
Y. Nikiforov et al., 2019 [47]	Yes	Prospective multicenter [47]	257 informative nodules	Not specified	Not specified	Bethesda III and IV focus [47]
Mark A Lupo et al., 2020 [48]	Yes	Retrospective blinded validation [48]	Not specified	Not specified	Not specified	Bethesda III, IV, V [48]
Busra Yaprak Bayrak et al., 2020 [49]	Yes	Retrospective cohort [49]	Not specified	Mean 1.9 cm [49]	Mean age 52.5 years, 85.2% female [49]	III: 4.2%, IV: 6.8% [49]
C. Nasr et al., 2022 [6]	Yes	Meta-analysis [6]	1976 patients across studies	Not specified	Not specified	Bethesda III and IV focus [6]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
M. Marina et al., 2021 [50]	Yes	Prospective [50]	91 Bethesda IV nodules	Mean 22 mm [50]	Mean age 54 years, 71% female [50]	All Bethesda IV [50]
S. Salameh et al., 2022 [51]	Yes	Retrospective chart review [51]	328 patients	GEA-positive: mean 2.1 cm (0.6-6.2) [51]	GEA-positive: mean age 46.8 years, 80% female [51]	III: 21.3%, IV: 34.7%, V: 16%, VI: 28% [51]
G. Fu et al., 2023 [52]	Yes	Prospective cohort [52]	580 patients	Not specified	Female: 75.2%, male: 24.8%; mean age 50 (F), 55 (M) [52]	ND: 18.2%, AUS: 30.3%, SFM: 18.2%, Malignant: 33.3% [52]
T. Grimmichova et al., 2022 [53]	Yes	Prospective cohort [53]	178 patients	Not specified	Not specified	II: 10.1%, III: 29.2%, IV: 20.3%, V: 21.9%, VI: 18.5% [53]
R. Sowmya et al., 2025 [54]	Yes	Prospective cross-sectional [54]	100 patients	Not specified	Not specified	V: 80%, VI: 100% malignancy [54]
Moraima Pagan et al., 2016 [55]	Yes	Prospective (VERA), Retrospective (CLIA) [55]	198 total	Not specified	Not specified	23% indeterminate (III, IV, V) [55]
Augustas Beiša et al., 2019 [56]	Yes	Prospective cohort [56]	219 patients	Not specified	Not specified	AUS/FLUS: 19.2%, FN/SFN: 57.5%, SUSP: 23.3% [56]
Qazi N. Ahmed et al., 2025 [57]	Yes	Prospective observational [57]	150 patients	Not specified	Not specified	II: 44%, IV: 16.7%, V and VI: 26% [57]
N. Şeker et al., 2021 [58]	Yes	Retrospective cohort [58]	425 patients	Not specified	Average age 46 years, 85.4% female [58]	I: 53, II: 193, III: 85, IV: 22, V: 61, VI: 11 [58]
M. Fawzy et al., 2021 [59]	Yes	Retrospective [59]	Not specified	Not specified	Not specified	III: 8.4%, IV: 15.3%, V: 6.3% [59]
M. Santos et al., 2020 [60]	Yes	Prospective multicenter [60]	435 patients, 440 nodules	Not specified	Not specified	III: 56.1%, IV: 43.9% [60]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
Tayyaba Ayaz et al., 2022 [9]	Yes	Retrospective [9]	137 patients	Median 3 cm [9]	Median age 55 years, 83% female [9]	III/IV: 44, V: 8, VI: 28 [9]
C. Durgun et al., 2023 [61]	Yes	Retrospective [61]	304 patients	Not specified	Not specified	I: 3.29%, II: 80.26%, III: 4.93%, IV: 4.27%, V: 3.95%, VI: 3.29% [61]
Karolina Skubisz et al., 2021 [62]	Yes	Retrospective cohort [62]	49 patients	Not specified	Not specified	All PTC patients [62]
A. Aras et al., 2024 [7]	Yes	Retrospective [7]	1229 patients	Not specified	Age 18-78, mean 45.22 years; 84% female [7]	III: 8%, IV: 12%, V: 33%, VI: 47% [7]
M. Quiriny et al., 2024 [63]	Yes	Prospective noninterventional [63]	Not specified	Not specified	Not specified	THY 1-6 categories [63]
Zhuochao Mao et al., 2023 [64]	Yes	Retrospective cohort [64]	3807 patients	Not specified; 45.9% between 5-10 mm [64]	75.4% female, 77.5% under 55 years [64]	I: 75, II: 116, III: 342, IV: 22, V: 665, VI: 2587 [64]
S. Mankovskaya et al., 2023 [65]	Yes	Prospective [65]	73 FNAB samples	Not specified	Not specified	IV: 19, VI: 41 [65]
M. Bak et al., 2015 [66]	Yes	Retrospective [66]	1384 patients	Not specified	Not specified	I: 15.9%, II: 73.1%, III: 4.2%, IV: 3.1%, V: 1.3%, VI: 2.4% [66]
Dr. Amrita Chowdhury et al., 2025 [67]	Yes	Retrospective cross-sectional [67]	54 patients	Not specified	45 females, 9 males; mean age 43.25 years [67]	Not detailed [67]
Dorota Whitmer et al., 2022 [68]	Yes	Retrospective [68]	8881 samples	Median 2.3 cm [68]	Higher GSC-S in males (6.9%) and <55 years (5.8%) [68]	III: 4.8% GSC-S, IV: 7.0% GSC-S [68]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
Hasan Athi et al., 2023 [69]	Yes	Retrospective [69]	146 patients	Not specified	Not specified	I: 33.3%, II: 14.5%, III: 18.5%, IV: 33.3%, V: 75.0%, VI: 100% [69]
M. Deniz et al., 2023 [70]	Yes	Retrospective cross-sectional [70]	475 participants	Mean 18.8 mm (0-65) [70]	383 women, 92 men; mean age 53.8 years [70]	B3: 17, B4: 5, B5: 11 [70]
Salih Celepli et al., 2022 [71]	Yes	Retrospective cohort [71]	780 patients	Not specified	Not specified	IV and V focus [71]
I. Rasa et al., 2022 [72]	Yes	Retrospective observational [72]	563 patients	Not specified	Not specified	FNA malignancy concordance: 77% [72]
D. Wylie et al., 2016 [73]	Yes	Retrospective [73]	235 initial, 42 second set	Not specified	Not specified	III and IV focus [73]
M. Livhits et al., 2018 [4]	Yes	Prospective parallel RCT [4]	149 nodules (54 in 2-4 cm range)	Median 2.0 cm (IQR 1.5-3.0) [4]	Median age 59 years, 81.9% female [4]	III: 89.9%, IV: 10.1% [4]
R. Kloos et al., 2016 [74]	Yes	Prospective multicenter [74]	1468 GEC benign patients	Not specified	Not specified	Bethesda III and IV focus [74]
A. Marcadis et al., 2019 [75]	Yes	Retrospective [75]	273 Bethesda III/IV	Not specified	Not specified	Bethesda III and IV [75]
M. Nikiforova et al., 2018 [76]	Yes	Analytical/retrospective tissue, [76]	228 tissue, 175 FNA	Not specified	Not specified	III: 84, IV: 74, V: 17 [76]
Guy Talmor et al., 2022 [77]	Yes	Retrospective [77]	230 patients	Not specified	Mean age 54.3 years, 73.0% female [77]	III: 103, IV: 64, V: 63 [77]
Trevor E. Angell et al., 2019 [78]	Yes	Blinded cohort [78]	943 blinded FNAs	Not specified	Not specified	III and IV nodules [78]
D. M. Carr et al., 2022 [79]	Yes	Retrospective [79]	36 with molecular testing	Not specified	Not specified	AUS/FLUS: 16, FN/SFN: 7, SM: 9, Malignant: 4 [79]

Study	Full text retrieved?	Study Type	Sample Size	Mean/Median Nodule Size	Patient Demographics	Bethesda Distribution
G. Randolph et al., 2022 [80]	Yes	Retrospective cohort [80]	211 validation cohort	Non-MTC: mean 2.6 cm (1.0-9.1) [80]	Non-MTC: mean age 52 years, 77% female [80]	Training: III: 24%, IV: 29%, V: 29%, VI: 19% [80]

Molecular Testing Platforms and Performance

The review identified studies utilizing various molecular testing platforms, with ThyroSeq (versions 2, 2.1, and 3) and Afirma (GEC, GSC, and GSC+XA) being the predominant platforms examined.

ThyroSeq Platform Studies

ThyroSeq platforms were utilized in multiple studies with varying performance metrics. The ThyroSeq v3 genomic classifier demonstrated sensitivity of 94% [3], specificity of 82% [3], negative predictive value (NPV) of 97% [3], and positive predictive value (PPV) of 66% [3] in a prospective multicenter validation study [3]. Another validation study reported similar performance with sensitivity of 90.9% [10], specificity of 92.1% [10], PPV of 76.9% [10], and NPV of 97.2% [10] for ThyroSeq v2.1 [10].

In a subsequent multicenter study, ThyroSeq v3 achieved a benign call rate of 61% [47], with 82% of histologically benign nodules correctly classified as negative [47], compared to 68% for Afirma GSC [47]. The false-negative rate was 3% [3], with all missed cancers being low-risk tumors [3].

Specific mutation detection capabilities varied by platform. ThyroSeq identified BRAF V600E in 20/76 cases (26%) [26] and RAS mutations in 9/76 cases (12%) [26]. When combined with cytology (Bethesda ≥ 5), BRAFV600E and TERT testing increased sensitivity to 80.0% [15] compared to cytology alone [15].

Afirma Platform Studies

The Afirma platforms demonstrated consistent high negative predictive values across studies. The Afirma Gene Expression Classifier (GEC) achieved sensitivity of 90% [74], specificity of 52% [74], and NPV $\geq 94\%$ [74] for Bethesda III and IV nodules [74]. The newer Genomic Sequencing Classifier (GSC) showed improved performance with real-world sensitivity of 97% [6], observed specificity of 88% [6], observed PPV of 65% [6], and NPV of 99% [6, 6].

The most recent Afirma GSC+XA platform demonstrated the highest performance metrics with sensitivity of 80.00% [29], specificity of 91.43% [29], PPV of 57.14% [29], NPV of 96.97% [29], and overall accuracy of 90.00% [29, 29]. This platform significantly outperformed older platforms in reducing unnecessary surgical interventions, with only 25% of GSC+XA patients undergoing surgery [29].

Afirma platforms demonstrated particularly high NPV values. In European experience, the GEC correctly identified all 16 malignancies as suspicious [13], though it had a false-negative rate of 37.5% with the GSC in oncocyctic-dominant cytology [13]. The benign call rate was 67% in real-world studies [6] compared to 54% in the validation study [6].

Platform Comparison

Direct head-to-head comparisons between ThyroSeq and Afirma were limited. In one parallel randomized trial, ThyroSeq v2 demonstrated higher specificity (91% vs. 66%) [4] and PPV (57% vs. 39%) [4] compared to Afirma GEC, with equivalent sensitivity of 100% for both platforms [4]. This translated to more patients avoiding surgery with ThyroSeq v2 (62%) compared to Afirma GEC (39%) [4].

A meta-analysis comparing molecular testing platforms found that ThyroSeq v3 had a higher benign call rate (61%) versus Afirma GSC (54%) [47], indicating ThyroSeq could potentially avoid more surgeries. However, observed PPV was higher in real-world GSC+XA studies (57.14%) [29] compared to ThyroSeq v3 validation studies (66%) [3].

The ThyGenX/ThyraMIR platform showed competitive performance with sensitivity of 47.37% [29], specificity of 88.30% [29], PPV of 45.00% [29], and NPV of 89.25% [29], though it had lower overall accuracy (81.42%) [29] compared to GSC+XA (90.00%) [29].

Bethesda Category and Malignancy Rates

Malignancy rates varied substantially by Bethesda category across studies. For Bethesda III (atypia of undetermined significance/follicular lesion of undetermined significance), reported malignancy rates ranged from 9.2% [7] to 47.36% [31], with most studies reporting rates between 20-35% [8, 42, 44]. One study focusing specifically on Bethesda III and IV nodules reported malignancy rates of 77.7% for Bethesda 3 and 76.4% for Bethesda 4 [8], though this represented a surgical cohort rather than all diagnosed nodules [8].

Bethesda IV (follicular neoplasm/suspicious for follicular neoplasm) showed malignancy rates ranging from 22.8% [7] to 69.2% [61], with a median around 30-40% [35, 46]. Bethesda V (suspicious for malignancy) demonstrated higher malignancy rates of 75.0% [69] to 100% [61, 66], as expected given the cytologic suspicion.

The concordance between Bethesda cytology and final histopathology varied. One study reported sensitivity of 66.6% [61] and specificity of 98.7% [61] for cytology alone, with accuracy of 93.5% [61]. Another study found cytology sensitivity of 80.1% [70] and specificity of 95.2% [70]. The positive predictive value for benign cytology ranged from 97-98% [66], while for malignant cytology it was 88-100% [66].

The integration of ultrasound findings with cytology improved diagnostic performance. Studies examining TI-RADS concordance with Bethesda categories showed that ACR TI-RADS 3 nodules had high concordance (94.6%) and negative predictive values (100.0%) [28], while TI-RADS 4 and 5 nodules showed lower positive predictive values (2.8% and 16.2%, respectively) [28] when correlated with Bethesda scores.

Final Histopathology and Cancer Prevalence

Cancer prevalence in surgical cohorts varied widely, reflecting different selection criteria and patient populations. In Bethesda III and IV nodules undergoing surgery, overall malignancy rates ranged from 22.5% [10] to 62.2% [12], with most studies reporting rates between 25-45% [3, 40, 53].

Papillary thyroid carcinoma (PTC) was the predominant malignancy identified across studies, representing 96.05% [36] to 98.2% [12] of malignant diagnoses in some cohorts [12, 36]. The distribution of PTC subtypes included classical variants (43-60.2%) [22, 62], follicular variants (17.8-33%) [22, 62], tall cell variants (18.8%) [22], and less common aggressive variants [22].

Noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) represented 4% [3] to 28.4% [30] of surgical specimens in studies that specifically reported this entity. The reclassification of NIFTP from malignant

to borderline significantly impacted the positive predictive values of molecular tests, with one study reporting PPV decreasing from 42.9% to 14.3% for ThyroSeq [24] when NIFTP was excluded from malignancy calculations.

Aggressive features in malignant nodules varied by molecular alteration pattern. BRAF-like nodules were more commonly associated with extrathyroidal extension (6.2% gross, 5.6% minimal) [5] and nodal metastases [22] compared to RAS-like or non-BRAF-non-RAS patterns [22]. In one cohort, 39.1% of oncocytic thyroid carcinomas (OTC) demonstrated aggressive features compared to 11.1% of follicular thyroid carcinomas (FTC) [25].

Gene expression alterations (GEAs) were associated with a 90.7% malignancy rate [51], with 48.5% having aggressive pathological features including extrathyroidal extension, lymph node metastasis, or aggressive histological subtypes [51]. Among GEA-positive nodules, 88.2% were PTC, 4.4% FTC, 2.9% Hürthle cell carcinoma, 2.9% NIFTP, and 1.5% poorly differentiated thyroid carcinoma [51].

Molecular-Histology Concordance

The concordance between preoperative molecular testing results and final histopathologic diagnosis showed considerable variation across platforms and study designs.

Overall Performance Metrics

For ThyroSeq platforms, sensitivity ranged from 87% [75] to 98.0% [76], with most studies reporting values between 90-95% [3, 10, 47]. Specificity ranged from 52% [75] to 92.1% [10], with higher values generally observed in more recent studies [3, 47]. The negative predictive value consistently exceeded 90% across studies, ranging from 93% [75] to 100% [4, 46].

For Afirma platforms, sensitivity ranged from 75.81% [29] for older GEC+GSC to 97% [6] for GSC in real-world studies. Specificity showed more variability, from 45.08% [29] to 91.43% [29], with the newest GSC+XA platform achieving the highest specificity [29]. Negative predictive values for Afirma consistently exceeded 94% [6, 74], with some studies reporting NPV as high as 99% [6].

The positive predictive value remained more variable for both platforms, influenced by cancer prevalence and NIFTP classification. ThyroSeq PPV ranged from 35% [75] to 76.9% [10], while Afirma PPV ranged from 30.72% [29] to 66.67% [29] depending on platform version and study population.

Performance by Bethesda Category

Molecular testing performance varied by Bethesda category. For Bethesda III nodules, one study reported 61.5% achieving optimal care with molecular testing versus 21.0% without testing [5]. In Bethesda IV nodules, optimal surgery was achieved in 75.6% with testing versus 32.1% without [5].

The benign call rate (BCR) varied by Bethesda category and platform. ThyroSeq v3 achieved a 61% BCR overall [47], classifying 82% of histologically benign nodules as negative [47]. Afirma GEC had a BCR of 43% [3], while ThyroidPrint showed a BCR of 67.1% [19], with higher rates for follicular lesions of undetermined significance (87%) compared to atypia of undetermined significance (58%) [19].

False-negative rates were generally low across platforms. ThyroSeq v3 had a 3% false-negative rate [3], with all missed cancers being intrathyroidal and low-risk [47]. The Afirma GSC showed a false-negative rate of 37.5% in oncocytic-dominant cytology [13], highlighting platform-specific limitations in certain tumor subtypes.

Mutation-Specific Performance

Specific genetic alterations showed varying concordance with malignancy. BRAF V600E mutation testing demonstrated 100% positive predictive value [11], with all BRAF-positive nodules confirmed as malignant on histology [11]. BRAF V600E was detected in 82.0% of papillary thyroid carcinomas [12], with subtype-specific performance: classical PTC 90.2% sensitivity versus follicular variant PTC 40.0% sensitivity [12].

RAS mutations showed different characteristics. Isolated RAS mutations were associated with low-risk thyroid cancer in 88.8% of tumor samples [40], though they did not definitively distinguish malignancy status [40]. The specificity of RAS variant detection alone was 88.8% [40], with sensitivity of 22.6% [40].

EIF1AX mutations conferred a 45.2% risk of malignancy or borderline diagnosis [26], but when coexisting with other driver mutations (RAS, TERT, TP53), the risk increased to 86% [26]. TSHR mutations demonstrated particularly high specificity, with 95% of TSHRpI568T variants classified as GSC-benign [68], indicating $\leq 4\%$ risk of malignancy [68].

Gene expression alterations (GEAs) showed 90.7% malignancy rate in positive cases [51], significantly higher than GEA-negative nodules (77.9%) [51]. Among GEA-positive malignancies, 48.5% demonstrated aggressive features [51], with BRAF V600E mutation significantly associated with aggressive GEA-positive nodules ($p < 0.001$) [51].

Bethesda-Histology Concordance

The concordance between Bethesda cytology categories and final histopathology demonstrated significant variability across institutions and nodule characteristics.

Malignancy Rates by Category

For Bethesda III nodules, malignancy rates ranged from 9.2% [7] to 77.7% [8] across studies. The wide variation reflects differences in study populations (all diagnosed nodules versus surgical cohorts), institutional practices in Bethesda classification, and geographic factors [36]. Studies with more conservative surgical selection reported lower malignancy rates (18.5% [69], 21.0% [5]), while surgical series showed higher rates (42.5% [44], 47.36% [31]).

Bethesda IV nodules demonstrated malignancy rates from 22.8% [7] to 76.4% [8]. Most studies reported rates between 30-40% [35, 46], aligning with published Bethesda system expectations. Subclassification of Bethesda III by architectural versus nuclear atypia showed different malignancy risks: architectural atypia 0% versus nuclear atypia 77-86% [33].

Bethesda V nodules showed consistently high malignancy rates of 75.0% [69] to 100% [57, 61], confirming the high predictive value of this category. However, 33.3% of Bethesda V patients with positive lymph nodes had previously undetected metastases [35], indicating limitations in preoperative staging.

Cytology Performance Metrics

Cytology alone demonstrated sensitivity ranging from 47.5% [69] to 95.4% [7], with specificity from 67% [7] to 98.7% [61]. The positive predictive value for benign cytology (Bethesda II) ranged from 97-98.25% [66], while for malignant cytology (Bethesda VI) it was 88.46-100% [66].

Agreement statistics between cytologic and histologic diagnosis showed variable concordance. One study reported an accuracy of 93.5% [61] for cytology, with Cohen's kappa indicating fair to moderate agreement [31]. The false-negative rate ranged from 17.2% [69] to 33.3% [2], primarily due to papillary microcarcinomas [58] and cases with Hashimoto's thyroiditis [58].

Upgrade and Downgrade Patterns

Repeat fine-needle aspiration in initially nondiagnostic cases (Bethesda I) showed reclassification with malignancy rates of 33.3% [69] to 42.7% [64] upon surgical resection. For Bethesda II (benign) cytology, the malignancy rate in surgical specimens was 6.5% [27] to 18.3% [42], substantially higher than expected.

Cases classified as Bethesda III underwent reclassification upon repeat FNA, with some upgrading to Bethesda V or VI (4 patients each) [44] and others downgrading to benign (2 patients) [44]. The presence of Hashimoto's thyroiditis or lymphocytic thyroiditis increased the rate of AUS/FLUS and suspicious cytology classifications [58], representing a source of false-positive cytologic interpretation.

Subtype-Specific Performance

Cytology performance varied by cancer subtype. Classical papillary carcinoma was detected with high accuracy [66], while follicular variant papillary carcinoma and follicular neoplasms were more challenging, requiring surgical excision for definitive diagnosis [30]. Oncocytic thyroid carcinoma (OTC) demonstrated 85.0% accuracy in preoperative cytology reporting oncocytic subtype [25], compared to 20.8% for follicular thyroid carcinoma (FTC) [25].

Minimally invasive follicular carcinomas were particularly difficult to distinguish from follicular adenomas on cytology [24], with NIFTP presenting similar challenges. One study found that 28.4% of surgical specimens were NIFTP [30], demonstrating the limitation of cytology in distinguishing these entities from invasive carcinoma or benign lesions.

Three-Way Concordance: Molecular Testing, Cytology, and Histopathology

Limited studies examined the concordance across all three diagnostic modalities—molecular testing, Bethesda cytology, and final histopathology—in an integrated manner.

Added Value of Molecular Testing

The addition of molecular testing to cytology significantly improved diagnostic accuracy. One study found that combining cytology (Bethesda ≥ 5) with BRAFV600E and TERT testing increased sensitivity to 80.0% [15] from 70% with cytology alone [15]. The specificity remained high at 100% [15], maintaining the accuracy while improving detection rates.

In patients with Bethesda III and IV nodules, molecular testing significantly improved the likelihood of optimal surgical management. Optimal surgical intervention was achieved in 67.5% of patients who underwent molecular testing compared with 25.1% in those who did not ($p < 0.001$) [5]. For Bethesda III nodules, 61.5% received optimal care with testing versus 21.0% without [5], while for Bethesda IV nodules, the rates were 75.6% versus 32.1% [5].

Molecular testing reduced unnecessary surgeries substantially. Studies reported that 99.0% of patients with benign molecular test results remained under surveillance without surgery [19], with only one patient undergoing surgery that revealed a follicular adenoma [19]. The benign call rate of 67.1% [19] translated to avoiding diagnostic surgery in approximately two-thirds of patients with indeterminate cytology.

Discordant Cases and Patterns

Discordance between molecular testing and histopathology occurred in specific patterns. False-positive molecular results included four benign follicular nodules, six follicular adenomas, and four oncocytic adenomas in one study

[19]. These cases represented molecular alterations that did not translate to malignant behavior, highlighting the distinction between molecular abnormalities and clinical malignancy.

False-negative molecular results, though rare (3%) [3], included intrathyroidal, low-risk tumors that would not have adversely affected patient outcomes if initially managed conservatively [47]. In one European cohort, the GSC failed to classify 6 out of 16 malignant nodules as suspicious [13], representing a 37.5% false-negative rate [13], particularly in oncocytic-dominant cytology [13].

The concordance between molecular testing and cytology showed interesting patterns. In a combined analysis, 63% concordance was observed between molecular diagnostics and [18F]FDG-PET/CT [17], with combined testing improving sensitivity to 95% [17] and specificity to 44% [17] for a double-negative test. For double-positive tests (positive molecular diagnostics plus positive imaging), specificity reached 86% [17].

Clinical Decision-Making Integration

The integration of molecular testing into clinical algorithms demonstrated measurable impact on surgical decisions. Studies comparing tested versus untested cohorts showed that 70.1% of suboptimally managed patients were from the group that did not undergo molecular testing [5]. Molecular testing influenced 92.3% of clinical decisions [19], primarily by enabling surveillance in lieu of surgery for benign results.

Platform-specific algorithms emerged from the data. For ThyroSeq positive results, specific groups of genetic alterations had cancer probabilities varying from 59% to 100% [3]. RAS-like mutations were associated with indolent behavior [5], while BRAF V600E and TERT promoter mutations indicated aggressive disease [5]. This stratification enabled more nuanced surgical planning beyond simple positive/negative classification.

For Afirma, the development of invasion and lymph node metastasis classifiers provided additional prognostic information. The invasion classifier achieved 99% NPV in validation [16], while the LNM classifier achieved 100% NPV [16], enabling identification of low-risk tumors that might be managed with less extensive surgery.

Subgroup Performance by Molecular Alteration

Specific molecular alterations demonstrated different concordance patterns with cytology and histology. BRAF V600E showed near-perfect concordance, with all positive cases confirmed as PTC on histology [11], though sensitivity varied by PTC subtype (90.2% for classical PTC versus 40.0% for follicular variant PTC) [12].

RAS mutations presented more complexity. While RAS-positive nodules had a 52.94% malignancy rate in one cohort [46], another study found that 88.8% of RAS-mutated tumors were low-risk cancers or NIFTPs [40]. The presence of multiple RAS family mutations (NRAS, HRAS, KRAS) across benign and malignant entities [50] made interpretation challenging without additional clinical or molecular context.

Gene expression alterations (GEAs) demonstrated high malignancy concordance (90.7%) [51] across Bethesda categories. When GEAs occurred with BRAF V600E, aggressive features were significantly more common ($p < 0.001$) [51]. However, GEAs alone without other mutations showed lower specificity, with some benign nodules harboring these alterations [51].

TSHR mutations illustrated platform-specific interpretation. While 96% of TSHRpI568T samples were classified as GSC-benign by Afirma [68], among the 4% classified as GSC-suspicious, 21% were malignant [68]. This suggested that while most TSHR mutations indicate benign disease, a subset may harbor malignancy requiring surgical evaluation.

Platform Comparison: ThyroSeq versus Afirma

Direct head-to-head comparisons between ThyroSeq and Afirma platforms were limited, with most studies using a single platform.

Comparative Performance Metrics

One randomized trial directly compared ThyroSeq v2 with Afirma GEC in parallel cohorts [4]. ThyroSeq v2 demonstrated higher specificity (91% vs. 66%) [4] and positive predictive value (57% vs. 39%) [4], while maintaining equivalent sensitivity of 100% for both platforms [4]. This translated to more patients avoiding surgery with ThyroSeq v2 (62%) compared to Afirma GEC (39%) [4].

A comparison of validation studies showed ThyroSeq v3 achieved 94% sensitivity [3], 82% specificity [3], 66% PPV [3], and 97% NPV [3], while Afirma GSC validation reported 91% sensitivity [6], 68% specificity [6], 47% PPV [6], and 96% NPV [6]. However, real-world Afirma GSC studies demonstrated improved performance with 97% sensitivity [6], 88% observed specificity [6], 65% observed PPV [6], and 99% NPV [6].

The benign call rate differed between platforms. ThyroSeq v3 achieved 61% BCR [47], correctly classifying 82% of histologically benign nodules as negative [47], compared to Afirma GSC's 54% BCR [47] with 68% of benign nodules correctly classified [47]. This suggests ThyroSeq may avoid more surgeries, though at potentially lower specificity.

Platform-Specific Strengths and Limitations

ThyroSeq platforms demonstrated particular strength in detecting specific mutations. The multi-gene panel identified point mutations in seven different genes and five types of gene fusions [10], providing detailed molecular characterization. This enabled risk stratification by specific genetic alterations, with cancer probabilities varying from 59% to 100% depending on the mutation profile [3].

ThyroSeq showed lower performance in detecting follicular variant papillary carcinoma (FVPTC), with only 40.0% sensitivity for BRAF V600E detection in FVPTC [12] compared to 90.2% in classical PTC [12]. This limitation extended to other low-frequency variants that may not be adequately represented in targeted sequencing panels [78].

Afirma platforms demonstrated consistently high negative predictive values across diverse populations. The gene expression classifier approach showed particular strength in ruling out malignancy, with NPV $\geq 94\%$ for Bethesda III/IV nodules [74] and up to 99% in real-world studies [6]. This high NPV enabled confident surveillance recommendations for negative results.

Afirma showed limitations in oncocytic neoplasms. Studies reported a low PPV of 20% for both ThyroSeq v3 and Afirma GSC in oncocytic aspirates [13], with the GSC demonstrating a 37.5% false-negative rate in oncocytic-dominant cytology [13]. This represents a significant limitation requiring alternative diagnostic approaches in this tumor subtype.

Cost-Effectiveness and Implementation

Practical implementation differences emerged from the studies. Afirma GSC+XA reduced the surgery rate to 25% of patients tested [29], representing a 75% reduction in unnecessary surgical intervention [29]. ThyroSeq v2 enabled 62% to avoid surgery [4], suggesting both platforms substantially reduce unnecessary procedures, though with different rates.

Cost comparisons showed Afirma testing was more expensive than some alternatives [43], though specific pricing was not disclosed in most studies. One study noted that determining which platform is more cost-effective is complex [29], depending on institutional factors and reimbursement structures.

Sample handling requirements differed between platforms. ThyroSeq v3 and Afirma GSC require either a new dedicated FNA or stored material from the original FNA [60], while platforms like mir-THYpe can utilize readily available cytology smear slides [37], potentially simplifying logistics and reducing the need for repeat procedures.

The technical success rate varied, with Afirma GSC achieving 91% informative results [4] and ThyroSeq achieving 90% informative results [3]. Failures typically resulted from insufficient cellular material or RNA quality [3, 48], emphasizing the importance of adequate sample collection and handling.

Emerging Alternative Platforms

Other molecular testing platforms demonstrated competitive performance. The ThyroidPrint platform achieved 91% sensitivity and 95% NPV [19] in validation studies, with a BCR of 67.1% [19]. This platform showed higher BCR for FLUS (87%) compared to AUS (58%) [19], indicating subtype-specific performance variation.

The ThyGenX/ThyraMIR combination demonstrated sensitivity of 47.37% [29], specificity of 88.30% [29], PPV of 45.00% [29], and NPV of 89.25% [29], with overall accuracy of 81.42% [29]. While showing lower sensitivity than ThyroSeq or Afirma, it maintained high specificity for ruling in malignancy.

MicroRNA-based classifiers showed promise with 89.3% sensitivity and 81.65% specificity [60] in real-world validation. These achieved similar NPV (95%) [60] to established platforms while offering the advantage of using existing cytology smears without requiring additional FNA passes [37].

Synthesis of Findings

The heterogeneity in reported concordance between molecular testing, cytology, and histopathology can be explained through several contextual factors rather than representing contradictory findings.

Population and Setting Effects

The malignancy prevalence in study populations significantly influenced predictive values. Studies with higher baseline cancer prevalence (40-60%) [40, 50] reported higher PPV for molecular tests compared to those with lower prevalence (20-30%) [3, 10], consistent with Bayesian principles. This variation reflects different patient selection—whether all diagnosed nodules or only surgical candidates were included [8, 9].

Geographic and institutional factors contributed to heterogeneity. Real-world Afirma GSC studies demonstrated better performance (sensitivity 97%, specificity 88%) [6] than the validation study (sensitivity 91%, specificity 68%) [6], attributed to increased surgical yield of true positives in clinical practice [6]. European experience with Afirma showed different performance characteristics than U.S. studies [13], possibly reflecting differences in thyroid disease prevalence or practice patterns.

The extent of surgery selection bias affected reported concordance. Studies assuming unoperated benign molecular results as true negatives reported higher NPV [6], while those with surgical follow-up of negative results showed somewhat lower NPV but more rigorous validation [3, 4]. This methodological difference explains apparent discrepancies between validation and real-world studies.

Tumor Biology and Molecular Context

The diagnostic challenge varies fundamentally by tumor type and molecular alteration pattern. BRAF V600E mutations show near-perfect concordance with papillary thyroid carcinoma [11], enabling definitive preoperative diagnosis. In contrast, RAS mutations occur in both benign (follicular adenomas) and malignant (follicular carcinoma, FVPTC) lesions [50], requiring consideration of additional factors like copy number alterations or gene expression patterns.

The NIFTP reclassification profoundly impacted molecular test performance metrics. Studies including NIFTP as malignant reported higher PPV [3] than those excluding it [24], with one study showing ThyroSeq PPV decreasing from 42.9% to 14.3% after NIFTP reclassification [24]. This reflects genuine biological uncertainty—NIFTP harbors molecular alterations associated with malignancy but lacks invasive features, representing an intermediate entity.

Oncocytic neoplasms demonstrate distinct molecular profiles with limited concordance to standard testing panels. Both ThyroSeq and Afirma showed reduced performance in oncocytic nodules (PPV 20% for both) [13], with Afirma GSC showing 37.5% false-negative rate in oncocytic-dominant cytology [13]. This indicates that oncocytic tumors require specialized diagnostic approaches or alternative molecular markers.

Methodological Quality and Study Design

Blinding and central pathology review significantly affected reported concordance. Prospective, double-blinded studies [3, 4] generally reported more conservative performance metrics than retrospective studies without explicit blinding [29, 46]. Central pathology review, particularly for NIFTP classification [75], standardized diagnoses and reduced inter-observer variability.

Sample size and statistical power influenced confidence intervals and reproducibility. Large multicenter studies with hundreds of samples [3, 47] provided more stable estimates than single-center studies with fewer cases [50, 79]. However, smaller studies sometimes identified important subgroup effects, such as age- or gender-specific performance differences [68].

The timing of molecular testing relative to surgery affected which patients received testing and underwent surgical confirmation [21]. Studies where molecular testing was reflexively performed on all indeterminate nodules [75] showed different characteristics than those where testing was selectively ordered based on clinical factors [46], introducing selection bias.

Verification bias represented a significant limitation across studies. Patients with benign molecular results predominantly underwent surveillance without surgical confirmation [19, 74], making true NPV difficult to determine definitively. The few studies with surgical follow-up of molecular negative results provided more rigorous validation [3, 10] but reflected selected populations willing to undergo diagnostic surgery despite reassuring molecular results.

Clinical Decision-Making Context

The added value of molecular testing beyond cytology alone varied by clinical scenario. For nodules with high-suspicion ultrasound features (ACR TI-RADS 5), molecular testing added limited incremental value [2], as surgery was indicated regardless. In contrast, for nodules with low-to-intermediate ultrasound risk (TI-RADS 3-4), molecular testing significantly refined risk stratification [33, 38].

The optimal surgical extent decision benefited from detailed molecular characterization beyond simple benign/malignant classification. Detection of aggressive mutations (BRAF V600E, TERT promoter) [5] guided decisions toward total thyroidectomy, while isolated RAS mutations suggested more conservative surgery might

suffice [5]. This nuanced approach required integrating molecular, cytologic, and imaging data rather than relying on any single modality.

Completion thyroidectomy rates varied based on initial molecular testing. Studies incorporating preoperative molecular testing showed more appropriate initial surgery selection [5], potentially reducing the need for subsequent completion procedures. However, definitive data on long-term completion thyroidectomy rates stratified by molecular test results remained limited across reviewed studies.

Research Gaps

Despite the substantial literature on molecular testing in thyroid nodules, several critical gaps remain:

Lack of nodule size-specific data: Very few studies specifically reported outcomes for the 2-4 cm nodule size range, despite this representing a critical decision point where both lobectomy and total thyroidectomy are viable options. Most studies included broad size ranges without stratified analysis, preventing definitive conclusions about molecular testing utility specifically for 2-4 cm nodules.

Limited head-to-head platform comparisons: Only one randomized study directly compared ThyroSeq and Afirma in parallel patient cohorts [4]. Most comparisons relied on indirect evidence from separate populations, limiting the ability to definitively establish which platform performs better in specific clinical contexts. No studies examined newer platform versions (ThyroSeq v4, Afirma GSC+XA) in head-to-head comparison.

Insufficient long-term outcomes data: Most studies focused on concordance with immediate surgical histopathology rather than long-term clinical outcomes. For patients with benign molecular results managed with surveillance, long-term follow-up data (5+ years) remained limited, preventing definitive assessment of false-negative rates and optimal surveillance strategies.

Inadequate economic analyses: Few studies rigorously examined cost-effectiveness incorporating not just test costs but also costs of unnecessary surgery, complications, and long-term thyroid hormone replacement. Platform-specific economic analyses accounting for different BCRs, specificity patterns, and clinical outcomes were notably absent.

Limited data on completion thyroidectomy: Studies rarely reported completion thyroidectomy rates after initial lobectomy stratified by preoperative molecular test results. This represents a critical gap, as one key value proposition of molecular testing is enabling appropriate initial surgery selection, potentially reducing the need for completion procedures and associated morbidity and costs.

Discussion

This systematic review of 80 studies demonstrates that molecular testing significantly improves risk stratification of thyroid nodules beyond cytology alone, though with important contextual limitations. The concordance between preoperative molecular testing results, Bethesda cytology scores, and final histopathology varies substantially based on platform choice, tumor biology, and clinical context.

ThyroSeq and Afirma platforms both achieve high negative predictive values ($\geq 94\%$), enabling confident surveillance recommendations for test-negative indeterminate nodules. However, positive predictive values remain more variable (35-77%), reflecting inherent biological heterogeneity in molecularly-altered thyroid nodules that include benign entities, NIFTPs, and low-risk cancers alongside clinically significant malignancies.

The added value of molecular testing is most pronounced in Bethesda III and IV nodules with low-to-intermediate ultrasound suspicion, where it can reduce unnecessary surgery by 60-75% while maintaining <5% false-negative

rates. In contrast, molecular testing adds limited value for high-suspicion nodules or those with benign molecular test results managed conservatively where surgery is indicated regardless of molecular results.

Critical gaps remain regarding platform performance specifically in 2-4 cm nodules, long-term outcomes of molecular test-negative surveillance, head-to-head platform comparisons, and economic evaluations. A large integrated database linking cytology, molecular testing, surgical decisions, and long-term outcomes would address these gaps and enable more precise, personalized surgical decision-making for patients with thyroid nodules in this critical size range.

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